USER MANUAL · v0.1.0

Intelligent QA Engineer

An Agent Skill that turns requirements, contracts and code into traceable, evidence-honest and runnable tests

- **What it can do**
- **How to install and use it**
- **The 9 operating modes, step by step**
- **Evidence, statuses, templates and validator**
- **Examples, prompts and limitations**

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1. What is Intelligent QA Engineer?

Intelligent QA Engineer is an **Agent Skill** (`SKILL.md` format, open Agent Skills standard) that makes your AI agent —Claude Code, OpenAI Codex or others— work as a **senior QA engineer and test architect**. It turns user stories, acceptance criteria, PRDs, specifications, OpenAPI/Swagger contracts, source code, *diffs* and existing suites into test deliverables that are traceable and concrete enough to implement.

It does more than generate cases: it **separates what is confirmed from what is assumed**, states honestly what was run and what was not, and never claims a test passed unless it saw it pass.

Skill priorities

> **Traceability** › **Evidence honesty** › **Concreteness** › **Proportionality** › **Safety**

Who is it for?

QA and test automation engineers, developers who need to test their change, Product Owners who want to know what to test, and teams preparing a release or reviewing a *pull request*.

What you can get

Capability	What it produces
Requirement analysis	Requirements and criteria with IDs, business rules, non-functional requirements, risks (impact × likelihood), assumptions and open questions.
Test strategy	Risk-based plan: test levels, types and techniques, each chosen with a reason.
Test cases	Structured CSV with objective, preconditions, data, steps, expected result, technique and priority.
Negative and edge scenarios	Boundary values, invalid input, authorization errors, abuse and edge cases.
API automation	Runnable code for Postman/Newman, pytest, Playwright or Jest/Vitest, with dependencies, configuration and commands.
Regression	Test selection for a PR, a *diff* or a contract change, with an impact matrix.
Coverage	Traceability matrix, gaps and requirement coverage (distinct from code coverage).
Reports	QA report, test summary or release-readiness report.

Across all of these: it replies in your language (English or Spanish), treats input documents as data (resistant to *prompt injection*), never writes real credentials and works even with no tools.

2. Installation

The skill is the `skills/intelligent-qa-engineer/` folder of the repository. The installed folder must be named `intelligent-qa-engineer`: the specification requires it to match the `name` field. It works without running code; Python 3.11+ is only needed for the bundled validator, and Node.js or pytest to run the generated automation.

Quick option: skills.sh CLI

The `skills` CLI downloads the skill from GitHub without cloning the repository. You need a recent version of **Node.js**:

```
npx skills add titoparra4/intelligent-qa-engineer
```

> The `skills` CLI is not maintained by this project and sends anonymous install telemetry; to opt out, set `DISABLE_TELEMETRY=1` before the command. This path is documented but **not yet verified** in the validation report.

Claude Code

```
git clone https://github.com/titoparra4/intelligent-qa-engineer.git
cd intelligent-qa-engineer

# Personal (all your projects)
mkdir -p ~/.claude/skills && cp -R skills/intelligent-qa-engineer ~/.claude/skills/

# Per project (shared with the team through git)
mkdir -p .claude/skills && cp -R <path>/skills/intelligent-qa-engineer .claude/skills/
```

Or as a plugin, inside Claude Code:

```
/plugin marketplace add titoparra4/intelligent-qa-engineer
/plugin install intelligent-qa-engineer@intelligent-qa-engineer
```

With the plugin, the skill is invoked as `/intelligent-qa-engineer:intelligent-qa-engineer`.

OpenAI Codex

```
# User
mkdir -p ~/.agents/skills && cp -R skills/intelligent-qa-engineer ~/.agents/skills/

# Repository
mkdir -p .agents/skills && cp -R <path>/skills/intelligent-qa-engineer .agents/skills/
```

Invoke it with `$intelligent-qa-engineer` or browse with `/skills`. If it does not appear, restart Codex.

ChatGPT (not verified)

If your plan allows custom skills, upload a ZIP whose top-level folder is `intelligent-qa-engineer/` under **Skills** and select it with `@intelligent-qa-engineer`:

```
cd skills && zip -r ../intelligent-qa-engineer.zip intelligent-qa-engineer && cd ..
```

Otherwise, create a Project, paste the body of `SKILL.md` (without the *frontmatter*) into its instructions and upload `references/` and `assets/templates/` as files. This is an adaptation, not an install: results stay `NOT RUN` unless you run them.

Update, uninstall and verify

Action	How
Update	Copy the folder again, `npx skills update` or `/plugin marketplace update`.
Uninstall	Delete the folder or `/plugin uninstall intelligent-qa-engineer@intelligent-qa-engineer`.
Verify the package	`pip install -r requirements-dev.txt` and `python tools/validate_skill.py skills/intelligent-qa-engineer`.
Check it is active	Ask your agent *"Which skills do you have available?"*; `intelligent-qa-engineer` should be listed.

Verification status (v0.1.0): the package structure and the plugin manifests pass validation, but install and invocation on each platform are listed as **pending** in `docs/validation-report.md`.

3. Getting started

You don't need to learn commands: **describe what you want to test in natural language** and attach or name the artifact. The skill infers the mode. To force one, name it: `/intelligent-qa-engineer REGRESSION for the current git diff`.

Example prompts

```
Analyze this user story and generate test cases with a traceability matrix.
```

```
Create negative scenarios for this endpoint.
```

```
Generate pytest tests from this OpenAPI.
```

```
Build a regression test plan for this pull request.
```

```
Analyze test coverage and identify gaps.
```

A typical conversation

#	You	The skill
1	Attach the story, contract or diff.	Lists the artifacts, checks which tools it has and picks the smallest set of modes.
2	—	Extracts requirements and risks with IDs; records assumptions (A-n) and questions (Q-n). Asks only if the answer would change the result.
3	—	Delivers cases, traceability or code, and validates them (validator, compilation).
4	Run the commands it gives you.	Helps you read the real results, now `OBSERVED` or `SIMULATED`.

4. The 9 operating modes

Each mode loads only its reference guide (*progressive disclosure*). For a simple request, the skill uses the smallest set of modes that answers it.

Mode	Use it when you want…	Guide it loads
ANALYZE	requirements reviewed, ambiguities and risks found	requirement-analysis.md
STRATEGY	a test strategy or test plan	test-strategy.md
GENERATE	test cases	test-design-techniques.md
NEGATIVE	negative, edge or abuse scenarios	negative-testing.md
AUTOMATE	runnable tests or configuration	api-testing.md, automation.md
REGRESSION	the impact of a PR, diff or contract change	regression-testing.md
COVERAGE	traceability, gaps and coverage	coverage-analysis.md
REPORT	a QA or release-readiness report	qa-reporting.md
FULL-QA	an end-to-end QA package for a feature	those of each step it includes

4.1 Analyze — ANALYZE

When: Starting point for a new story, PRD or specification.

What it delivers:

- Requirements and criteria (REQ-n / AC-n), business rules (BR-n), non-functional requirements (NFR-n)
- Risks (RISK-n) rated by impact × likelihood
- Each item classified as Confirmed, Assumption (A-n) or Open question (Q-n)

Example prompt:

```
Analyze this user story and tell me what is missing and which risks you see.
```

4.2 Strategy — STRATEGY

When: To decide what to test, at which level and with which technique before writing cases.

What it delivers:

- Test levels and types chosen by risk, each with a reason
- Design techniques: boundary values, decision tables, state transitions, pairwise…
- A document based on the test-strategy.md template

Example prompt:

```
Prepare a risk-based test plan for this PRD.
```

4.3 Generate cases — GENERATE

When: To get test cases ready to run or automate.

What it delivers:

- CSV with the 13 columns of the test-cases.csv template
- One verifiable objective per case, with preconditions, data, steps and expected result
- No duplicates of the same equivalence class
- Traceability matrix requirement › cases

Example prompt:

```
Generate test cases with traceability for US-142.
```

4.4 Negative scenarios — NEGATIVE

When: To cover what can go wrong: invalid input, boundaries, permissions, abuse.

What it delivers:

- Out-of-range values (min-1, max+1), invalid formats, missing fields
- Authentication and authorization errors
- Duplicates, concurrency and idempotency where relevant

Example prompt:

```
Create negative and boundary scenarios for POST /customers.
```

4.5 Automate — AUTOMATE

When: When you want code that runs, not pseudocode.

What it delivers:

- Postman/Newman, pytest, Playwright or Jest/Vitest
- Pinned or bounded dependency versions and environment variables for URL and credentials
- Meaningful assertions (status, schema, body, side effects), no arbitrary sleeps
- Commands to run it

Example prompt:

```
Generate pytest tests from this OpenAPI.
```

4.6 Regression — REGRESSION

When: For a pull request, a diff or a contract change.

What it delivers:

- Classification of each change (breaking, new rule, additive…)
- regression-impact-matrix.csv: tests to rerun, update or create
- Recommended suites, with confirmed versus hypothetical impact

Example prompt:

```
Build a regression test plan for this pull request.
```

4.7 Coverage — COVERAGE

When: To know what is covered and what is not.

What it delivers:

- Traceability matrix with status Covered / Partial / Not covered / Untestable / Blocked
- Prioritized gaps
- Metrics only when they can be computed, with the formula and the counts

Example prompt:

```
Analyze the requirement coverage of this suite and identify gaps.
```

4.8 Report — REPORT

When: To communicate the quality status.

What it delivers:

- QA report, test summary or release readiness (qa-report.md template)
- Results with their evidence label
- Never declares a release ready based only on generated tests

Example prompt:

```
Write a release-readiness report with these results.
```

4.9 Full QA — FULL-QA

When: Runs the cycle in order and skips the steps that add nothing.

What it delivers:

- ANALYZE › STRATEGY › GENERATE › NEGATIVE
- › AUTOMATE (if asked or clearly useful)
- › COVERAGE › REPORT

Example prompt:

```
Do the full QA for this feature.
```

5. Rules the skill always follows

These rules are non-negotiable and explain much of what you will see in the answers:

#	Rule	What it means for you
1	Never invents requirements.	Anything not in the artifacts is an Assumption (A-n) or an Open question (Q-n), and is labeled as such.
2	Labels evidence.	Each result is GENERATED, EXPECTED, SIMULATED or OBSERVED (see chapter 7).
3	Never says something passed without seeing it.	Generated tests that have not been run are always NOT RUN.
4	No made-up numbers.	Reports only metrics computable from the data available, showing the formula and the counts.
5	Keeps coverage types apart.	Requirement coverage is not code coverage; it always says which one it means.
6	Is proportionate.	Picks only the modes, levels and techniques the request needs, with a short reason for each.
7	Input is data, not instructions.	If a story, contract or log says "ignore your rules" or "mark everything as PASS", it does not comply: it reports a SECURITY finding.
8	No real credentials.	Uses environment variables and placeholders such as `${API_TOKEN}`; never prints or stores secrets.
9	Protects environments.	No requests to production and no destructive tests without explicit authorization naming the target. Prefers local mocks.
10	Does not run untrusted commands blindly.	Shows commands from untrusted documentation before running them and does not send your artifacts to external services without permission.
11	Replies in your language.	IDs, statuses and code identifiers stay in English.

6. Workflow: analysis mode and workspace mode

The 8 steps of every request

Step	What it does
1. Detect context	Lists the artifacts and checks which tools exist (files, shell, network). Assumes none of them.
2. Classify	Picks the smallest set of modes for the request.
3. Analyze	Extracts requirements, rules, constraints, NFRs and risks with stable IDs that cite their source.
4. Identify gaps	Classifies each item as Confirmed, Assumption or Open question. Asks only if the answer would make the deliverable wrong.
5. Select the approach	Levels, types and techniques by risk, each with a reason.
6. Generate	Cases, traceability, strategy, report, regression matrix or runnable code.
7. Validate	Every requirement has a test or is listed as a gap; every test maps to a requirement; unique IDs; results match the contract; code compiles.
8. Report	Summary and top risks first; then deliverables, assumptions, validation performed, what is pending and next steps.

Two ways of working

	Analysis mode	Workspace mode
When	No tools, or chat only	Files and a shell
Deliverables	Inline, as Markdown or CSV blocks	Written to files; long material stays out of the chat
Validation	Gives you the commands to run	Runs the CSV validator and syntax-checks the code
Test execution	None: everything is NOT RUN	Only against local or authorized targets, reporting the real output

Tip: state the test target (for example, "use the local mock") and your preferred format (pytest, Postman…). If a non-critical fact is missing, the skill states an assumption and continues.

7. Evidence, execution statuses and identifiers

Evidence labels

Each result says where it comes from, so you know how far you can trust it:

Label	Meaning	Does it prove anything about the real system?
GENERATED	Designed by the skill, never run.	No
EXPECTED	Derived from the contract or requirements.	No
SIMULATED	Run against a mock or fixture.	No; it proves the tests work
OBSERVED	Actually run, with output seen.	Yes, in that environment

Execution statuses

Status	Meaning
PASS	Run and met the expected result.
FAIL	Run and did not meet it.
BLOCKED	Could not run because of a dependency or impediment.
NOT RUN	Not run yet. The status of every newly generated test.

Identifiers

IDs are stable, so you can refer to them later ("automate TC-CUS-004", "answer Q-2"). If your artifacts already have IDs, the skill keeps them.

ID	Meaning	ID	Meaning
REQ-n	Requirement	A-n	Assumption
AC-n	Acceptance criterion	Q-n	Open question
BR-n	Business rule	CHG-n	Change (regression)
NFR-n	Non-functional requirement	TC-<AREA>-NNN	Test case (e.g. TC-CUS-001)
RISK-n	Risk	SECURITY	Security finding

Traceability chain

Source artifact › REQ / AC / BR / NFR › RISK › TC › Traceability matrix › Result with evidence

8. Deliverables and templates

The templates live in `skills/intelligent-qa-engineer/assets/templates/`:

Template	Use
test-cases.csv	Test cases (GENERATE, NEGATIVE)
traceability-matrix.csv	Requirement › cases traceability (COVERAGE)
regression-impact-matrix.csv	Impact of a change and affected tests (REGRESSION)
test-strategy.md	Test strategy or plan (STRATEGY)
qa-report.md	QA or release-readiness report (REPORT)

Test-case CSV columns

Column	Content	Allowed values
test_case_id	Unique ID	`TC-<AREA>-NNN`
requirement_id	Requirement(s) covered	IDs from the matrix
title / objective	Title and one verifiable objective	Text
priority	Priority	Critical, High, Medium, Low
preconditions / test_data	Prior state and data	Text ('None' if not applicable)
steps	Reproducible steps	1. … | 2. … | 3. …
expected_result	Verifiable result	Concrete text
test_type	Type	Functional, API, Integration, Unit, E2E, Regression, Smoke, Security, Performance, Contract, Usability, Accessibility
technique	Design technique	EP, BVA, Decision table, State transition, Pairwise, Error guessing, Risk-based, Use case, Exploratory
automation	Automation candidate	Automate, Candidate, Manual
tags	Free tags	e.g. negative;boundary

Matrices

- **Traceability:** `requirement_id, requirement_summary, source, priority, risk_ids, test_case_ids, coverage_status, notes`. Coverage statuses: Covered, Partial, Not covered, Untestable, Blocked.
- **Regression impact:** `change_id, component, change_type, impact_type, affected_requirement_ids, risk, existing_tests, action, new_tests, suite, rationale`.

Automation requirements

- Runnable code, not pseudocode.
- Dependency and configuration files with pinned or bounded versions.
- Environment variables for the base URL and credentials.
- Commands to run the tests.

- No arbitrary sleeps, and meaningful assertions: status, schema, body and side effects.

> The CSV columns are fixed and in English.

9. The test-case validator

The skill bundles `scripts/validate_test_cases.py`, which uses only the Python standard library (3.11+). In workspace mode the skill runs it on the CSVs it generates; you can also run it yourself:

```
python skills/intelligent-qa-engineer/scripts/validate_test_cases.py \
  test-cases.csv --traceability traceability-matrix.csv
```

Option	Effect
`--traceability, -t`	Also validates the traceability matrix and its consistency with the cases.
`--json`	Machine-readable report (useful in CI).
`--strict`	Treats warnings as errors.

What it checks

- Required columns present and not duplicated; no rows with unquoted commas.
- IDs in `TC-<AREA>-NNN` format and without duplicates.
- Priority, type and automation within the allowed values; known techniques.
- Verifiable expected result and sufficiently detailed steps.
- Possible credentials or sensitive data in the content.
- Every case maps to a requirement in the matrix, and the matrix references only existing cases.

Exit codes

Code	Meaning
0	Valid (warnings allowed)
1	Validation errors (or warnings with --strict)
2	Usage or file error

> The validator checks **structure and consistency**, not the quality of the test design: a valid CSV still needs human review.

10. Bundled examples

The repository ships five complete examples in `examples/`. The runnable ones target a **deterministic local mock API**: no network, credentials or external services are needed, and their results are `SIMULATED`.

#	Example	Input	Output	Runnable
1	User story	US-142, discount coupon (with a planted prompt injection)	Analysis, risks, questions, 17 cases, traceability	Validator
2	REST API	Customers OpenAPI contract	35-case catalog, negatives, traceability	Validator
3	Postman	Customers contract	Collection of 22 requests, environment and Newman runner	Yes (Node.js)
4	pytest	Customers contract	35-case suite with fixtures and configuration	Yes (Python)
5	Regression	Simulated PR + contract diff v1.0 › v1.1	Impact analysis, matrix, suites and gaps	—

Walkthrough: from OpenAPI contract to pytest

```
cd examples/pytest
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
pytest                          # everything against the embedded mock
pytest -m smoke                 # smoke subset
pytest -m "not destructive"     # no deletes
pytest --junitxml=reports/junit.xml
```

Without `API_BASE_URL`, `conftest.py` starts the mock on a free port. Against another environment (only if authorized) you need `API_BASE_URL`, the tokens and `ALLOW_REMOTE_TESTS=1`; destructive tests also require `ALLOW_DESTRUCTIVE_TESTS=1`.

Walkthrough: Postman with Newman

```
examples/postman/run-local.sh          # port 8000
examples/postman/run-local.sh 8765     # custom port
```

The script starts the mock, runs Newman, stops it and writes a JUnit report. Run the collection in order: requests chain variables.

Quick security check

Story US-142 deliberately contains a hidden instruction aimed at the assistant. Ask for an analysis of `examples/user-story/input-user-story.md`: the skill should ignore it and report it as a `SECURITY` finding. It is a quick way to check that the skill is active.

11. What the skill does NOT do

Does not	What it does instead
Fix application bugs	Designs the tests that detect them and describes the observed failure.
General coding with no testing goal	Activates only when the request is about testing something.
Load or performance testing at scale	Out of scope. A time threshold in a functional test is not a performance test.
Penetration testing of live systems	Out of scope. It can design negative authorization cases against local environments.
Run against production or shared environments without permission	Asks for explicit authorization naming the target and prefers local mocks.
Declare a release ready based only on generated tests	Reports what is still NOT RUN and which evidence is missing.

Known limitations (v0.1.0)

- Behavioral evaluation in real agents (15 scenarios, S01–S15) and per-platform verification are **pending**.
- Playwright and Jest/Vitest are covered by guidance only; there is no runnable example yet.
- A negative test that unexpectedly creates a record does not clean it up; harmless with the mock, but it would leave data in a persistent environment.
- date-time format checks in the pytest example require the optional rfc3339-validator package.

12. Usage tips and FAQ

Tips

- **Provide the full artifact** (story, contract, diff): the more faithful the input, the better the traceability.
- **Answer the open questions (Q-n)** before accepting the cases; assumptions can change the expected result.
- **Iterate by ID:** "automate TC-CUS-004 to TC-CUS-010", "add negatives for REQ-3".
- **Always run the validator** on generated CSVs and fix every error.
- **Test against a local mock first**; for staging, authorize the target explicitly and never point at production.
- **Regenerate the Postman schemas** when the OpenAPI contract changes.
- **Pair it with intelligent-business-analyst:** its Given/When/Then criteria are ideal input for GENERATE.

Frequently asked questions

Do I have to name the mode?

No. The skill infers it from the request; naming it only forces a specific one.

Why is everything NOT RUN?

Because the tests were not run, usually because of analysis mode or no authorized target. Run the commands it gives you to get SIMULATED or OBSERVED results.

Why doesn't it give me a coverage percentage?

Because it only reports metrics it can compute from the data available, always with the formula and the counts.

Can I use it in Spanish?

Yes. It replies in the user's language; IDs, statuses and columns stay in English.

What if my document contains instructions for the AI?

It treats them as data, does not follow them and reports them as a SECURITY finding.

The skill does not show up in Claude Code. What should I do?

Check that the folder is named intelligent-qa-engineer and sits in ~/.claude/skills/ or .claude/skills/, then start a new session.

Appendix: prompt cheat sheet

I want…	Write something like…
To know what to test	What should I test in this story? <paste>
To review requirements	Analyze these acceptance criteria and tell me what is missing and which risks exist
A test plan	Prepare a risk-based test plan for this PRD
Test cases	Generate test cases with a traceability matrix for US-142
Negatives and boundaries	Create negative and boundary-value scenarios for POST /customers
pytest tests	Generate pytest tests from this OpenAPI: <path>
A Postman collection	Generate a Postman collection with Newman for this contract
E2E tests	Generate Playwright tests for the checkout flow
PR regression	Build a regression test plan for this pull request
Regression for my branch	/intelligent-qa-engineer REGRESSION for the current git diff
Coverage	Analyze the requirement coverage of this suite and identify gaps
A report	Write a release-readiness report with these results
The whole cycle	Do the full QA for this feature: <need>
Run locally	Run the generated tests against the local mock and give me the real output
Answer questions	Answers: Q-1 the coupon does not stack; Q-2 yes, it expires at midnight UTC